

Infinite horizon optimal growth model

Consider the same environment as before, with one key difference: $T = \infty$

$\Rightarrow$ The Social Planner / Robinson crasse / agent's (depending on interpretation) problem is:

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} \leq f(k_t), \quad t \geq 0 \quad (t=1, 2, \dots) \\ & k_{t+1} \geq 0, \quad t \geq 0 \\ & k_0 \text{ is given} \\ & c_t \geq 0 \quad (\text{unnecessary if } u'(0) \rightarrow \infty) \end{aligned}$$

Remark: $\beta \in (0, 1)$ is crucial!

If $\beta \geq 1 \Rightarrow$ the objective function is infinite $\Rightarrow$ the problem is not well-defined

Trivially, $c_t + k_{t+1} \equiv f(k_t)$ (not optimal to waste resources)

$\Rightarrow$ Rewrite (*) as

$$\begin{aligned} \max_{\{k_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(f(k_t) - k_{t+1}) \\ \text{s.t.} \quad & 0 \leq k_{t+1} \leq f(k_t), \quad t \geq 0 \\ & k_0 \text{ is given} \end{aligned}$$

(SP)
for
"sequential
problem"

If $f(0)=0$ and $u'(0)=-\infty \Rightarrow$ the constraints are not binding $\Rightarrow$ the problem is unconstrained, just take (FOC) w.r.t. (k_{t+1}) :

$$-\beta^t u'(f(k_t) - k_{t+1}) + \beta^{t+1} \cdot f'(k_{t+1}) \cdot u'(f(k_{t+1}) - k_{t+2}) = 0$$

$\Rightarrow$ get a familiar second-order difference equation:

(EE)

$$u'(f(k_t) - k_{t+1}) = \beta f'(k_{t+1}) \cdot u'(f(k_{t+1}) - k_{t+2}), t \geq 1$$

k_0 is given

Notice, however, that it has only one terminal condition (k_0 is given) $\Rightarrow$

$\Rightarrow$ (EE) has an infinite # of solutions (continuum):

$$(\forall k_1) : \exists k_2 \Rightarrow \exists k_3 \dots$$

undetermined

(because $\nexists k_{T+1} = 0$)

Under certain conditions on $u(\cdot)$ and $f(\cdot)$, the sequence of $\{k_{t+1}\}_{t=0}^{\infty}$ solves (SP) if and only if $\bullet \{k_{t+1}\}_{t=0}^{\infty}$ solves (EE)
 $\bullet$ satisfies TRANSVERSALITY condition:

(TC)

$$\lim_{t \rightarrow \infty} \beta^t u'(f(k_t) - k_{t+1}) \circ f'(k_t) \cdot k_t = 0$$

Pl. at an extra unit of k_t

PV of one extra unit of k_t

PV of extra utility generated by k_t
in period t .

For brevity, denote $F(x_t, x_{t+1}) = U'(f(x_t) - x_{t+1})$
the proof holds for more general case.

Theorem:

Suppose $F(\cdot, \cdot)$ is continuously differentiable,
concave, $\uparrow$ in the first argument, $\downarrow$ in the second
argument and bounded.

Then $\{x_t\}_{t=0}^{\infty}$ solves (SP) if and only if
 $\{x_t\}_{t=0}^{\infty}$ satisfies (EE) & (TC):

$$(EE): \quad F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0$$

$$(TC): \quad \lim_{t \rightarrow \infty} \beta^t F_1(x_t, x_{t+1}) \cdot x_t = 0$$

Proof: ① Sufficiency
② necessity

1. Sufficiency

Let $\{x_0^*, x_1^*, \dots, x_t^*, \dots\}$ satisfy (EE) and (TC)

Need to show that $\forall \{x_t\}_{t=0}^{\infty} \neq \{x_t^*\}_{t=0}^{\infty}$,

$$\sum_{t=0}^{\infty} \beta^t F(x_t^*, x_{t+1}^*) \geq \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1})$$

need to show

$$\mathcal{D} = \lim_{T \rightarrow \infty} \sum_{t=0}^T \beta^t [F(x_t^*, x_{t+1}^*) - F(x_t, x_{t+1})] \geq 0$$

$F(x, y)$ - concave $\Rightarrow$

$$\Rightarrow F(x^*, y^*) - F(x, y) \geq F_x(x^*, y^*)(x^* - x) + F_y(x^*, y^*)(y^* - y)$$

$$\Rightarrow \mathcal{D} \geq \lim_{T \rightarrow \infty} \sum_{t=0}^T \beta^t \cdot [F_1(x_t^*, x_{t+1}^*) \cdot (x_t^* - x_t) + F_2(x_t^*, x_{t+1}^*)(x_{t+1}^* - x_{t+1})]$$

$$= \lim_{T \rightarrow \infty} F_1(x_0^*, x_1^*)(x_0^* - x_0) + F_2(x_0^*, x_1^*)(x_1^* - x_1) +$$

Since x_0 is given $\Rightarrow 0$ by (EE)

$$+ \beta F_1(x_1^*, x_2^*)(x_1^* - x_1) + \beta F_2(x_1^*, x_2^*)(x_2^* - x_2) + \dots$$

$$+ \beta^T F_1(x_T^*, x_{T+1}^*)(x_T^* - x_T) + \beta^T F_2(x_T^*, x_{T+1}^*)(x_{T+1}^* - x_{T+1})$$

$$= \lim_{T \rightarrow \infty} \beta^T F_2(x_T^*, x_{T+1}^*) x_{T+1}^* - \lim_{T \rightarrow \infty} \beta^T F_2(x_T^*, x_{T+1}^*) \cdot x_{T+1} \geq 0$$

$$= | \text{by (EE)} | = \geq 0$$

$$= - \lim_{T \rightarrow \infty} \beta^{T+1} F_1(x_{T+1}^*, x_{T+2}^*) \cdot x_{T+1}^* = 0 \text{ by (TC)}$$

Thus if $\{x_t^*\}_{t=0}^{\infty}$ satisfies (EE) + (TC) $\Rightarrow \{x_t^*\}_{t=0}^{\infty}$ solves (SP)

2. Necessity

Suppose that $\{x_t^*\}_{t=0}^\infty$ solves (SP) $\Rightarrow$

$\Rightarrow$ any other $\{x_t\}_{t=0}^\infty$ delivers less life time utility.

In particular, consider $\{x_t\} = \{x_0^*, x_1^*, \dots, x_T^*, \lambda x_{T+1}^*, \lambda x_{T+2}^*, \dots\}$

This is feasible ($x_t \geq 0$)

$\lambda \in [0, 1]$

$$\Rightarrow \sum_{t=0}^{\infty} \beta^t [F(x_t^*, x_{t+1}^*) - F(x_t, x_{t+1})] \geq 0$$

$$\parallel \beta^T [F(x_T^*, x_{T+1}^*) - F(x_T^*, \lambda x_{T+1}^*)] + \sum_{t=T+1}^{\infty} \beta^t [F(x_t^*, x_{t+1}^*) - F(\lambda x_t^*, \lambda x_{t+1}^*)] \geq 0$$

$$= \beta^T \frac{F(x_T^*, x_{T+1}^*) - F(x_T^*, \lambda x_{T+1}^*)}{1 - \lambda} \leq \sum_{t=T+1}^{\infty} \beta^t \frac{F(x_t^*, x_{t+1}^*) - F(\lambda x_t^*, \lambda x_{t+1}^*)}{1 - \lambda}$$

take $\lim_{\lambda \rightarrow 1}$

F -concave in each arg. $\Rightarrow$

$$\frac{F(x, y) - F(\lambda x, \lambda y)}{1 - \lambda} \leq F(x, y) - F(0, 0)$$

$$- \beta^T \lim_{\lambda \rightarrow 1} \frac{F(x_T^*, x_{T+1}^*) - F(x_T^*, \lambda x_{T+1}^*)}{(1 - \lambda) x_{T+1}^*} \leq \sum_{t=T+1}^{\infty} \beta^t [F(x_t^*, x_{t+1}^*) - F(0, 0)]$$

$$= \beta^T F_2(x_T^*, x_{T+1}^*) \cdot x_{T+1}^* \leq \sum_{t=T+1}^{\infty} \beta^t [F(x_t^*, x_{t+1}^*) - F(0, 0)]$$

by (EE)

$$\beta^{T+1} F_1(x_{T+1}^*, x_{T+2}^*) \cdot x_{T+1}^* \leq \text{---} \parallel \text{---}$$

$$\lim_{T \rightarrow \infty} \beta^T F_1(x_T^*, x_{T+1}^*) \cdot x_{T+1}^* \leq \lim_{T \rightarrow \infty} \sum_{t=T+1}^{\infty} \beta^t \cdot (\text{bounded}) = 0$$

$\Rightarrow$ (TC) must hold!

